

Lesson 18

Geometric Probabilities

Learning Objectives

In this lesson, we will study geometric probability and how probability can be computed using geometric measures such as length, area, and volume. We will learn how to model random events using intervals, regions, and geometric shapes, and how to calculate probability by comparing successful regions to total possible regions. By applying these ideas, we will solve competition problems involving random points, random cuts, and geometric regions commonly seen in AMC and math contest settings.

© Concept 1: Definition and Formula of Geometric Probability

Geometric probability is an approach to understanding the problem of infinite outcomes by measuring the number of outcomes geometrically, in terms of length, area, or volume, based on the assumption that each possible outcome is equally likely to occur.

The probability of event A is $P(A) = \frac{m}{n}$, where m is the length, area, or volume of event A and n is the total length, area, or volume of the all possible outcomes of the experiment.

Warm Up

Which of the following represents a geometry probability case?

- A. Choose a and b from 1, 2, 3, 4 without repetition and determine the probability that equation $x^2 + 2ax + b = 0$ has a real root.
- B. Suppose that a and b satisfy $|a| \leq 2$ and $|b| \leq 3$. Find the probability $x^2 + 2ax + b = 0$ has a real root.
- C. Select two persons from A , B , and C to form a team. Find the probability that A is in the team.
- D. Determine the probability that Jack and Austen share the same birthday.

⊙ Concept 2: Probability Using Lengths

Probability Using Lengths:

$$P(\text{Desired Event}) = \frac{\text{Length of successful region}}{\text{Length of possible region}}.$$

Math Exploration 1

- 1 A line segment of length 1 is randomly cut into two parts. What is the probability that both parts are at least $\frac{1}{3}$ in length?
- A. $\frac{1}{6}$ B. $\frac{1}{4}$ C. $\frac{1}{3}$ D. $\frac{1}{2}$
- 2 Given that x is an arbitrary point on the interval $[-5, 5]$, then the probability that x satisfies $x^2 - x - 2 \geq 0$ is _____.

- 3 Chloé chooses a real number uniformly at random from the interval $[0, 2017]$ independently, Laurent chooses a real number uniformly at random from the interval $[0, 4034]$. What is the probability that Laurent's number is greater than Chloé's number? (Assume they cannot be equal).

(2017 AMC 10A Problem, Question #15)

- A. $\frac{1}{2}$ B. $\frac{2}{3}$ C. $\frac{3}{4}$ D. $\frac{5}{6}$ E. $\frac{7}{8}$

- 4 Points A and B lie on a circle with radius 2, and the length of segment AB is 2. A point P is randomly chosen on the circle.

What is the probability that triangle $\triangle PAB$ is an acute triangle?

- A. $\frac{1}{3}$ B. $\frac{1}{4}$ C. $\frac{1}{6}$ D. $\frac{2}{5}$ E. $\frac{1}{2}$

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© Concept 3: Probability Using Areas

$$P(\text{success}) = \frac{P(\text{Area of successful outcomes region})}{P(\text{Area of total outcomes region})}.$$

Math Exploration 2

- 1 Cut a 1-meter-long stick into three pieces, then the probability that each piece of stick is less than 0.5 m is _____ .
- 2 Take three points on a circle to make a triangle, then what is the probability that this triangle is acute?

Math Exploration 3

- 1 Two real numbers x and y are randomly chosen from the interval $[-2, 2]$.

What is the probability of the event $A : x + y \geq 1$? _____

- 2 Real numbers x and y are chosen independently and uniformly at random from the interval $[0, 1]$. Which of the following numbers is closest to the probability that x, y , and 1 are the side lengths of an obtuse triangle? (2018 AMC 10B Problems, Question #22)

A. 0.21 B. 0.25 C. 0.29 D. 0.50 E. 0.79

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Math Exploration 4

- 1 A point is chosen at random within the square in the coordinate plane whose vertices are $(0, 0)$, $(2020, 0)$, $(2020, 2020)$, and $(0, 2020)$. The probability that the point is within d units of a lattice point is $\frac{1}{2}$. (A point (x, y) is a lattice point if x and y are both integers.) What is d to the nearest tenth? (2020 AMC 10A Problems, Question #16)

A. 0.3 B. 0.4 C. 0.5 D. 0.6 E. 0.7

- 2 Two concentric circles have radii 1 and 2. Two points on the outer circle are chosen independently and uniformly at random. What is the probability that the chord joining the two points intersects the inner circle? (2014 AMC 10B Problems, Question #19)

A. $\frac{1}{6}$ B. $\frac{1}{4}$ C. $\frac{2 - \sqrt{2}}{2}$ D. $\frac{1}{3}$ E. $\frac{1}{2}$

- 1 A real number x is randomly selected from the interval $[-3, 4]$. What is the probability that $|x| \leq 1$?

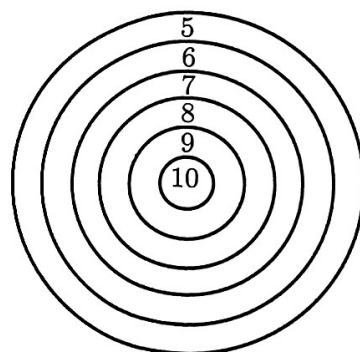
A. $\frac{1}{7}$

B. $\frac{6}{7}$

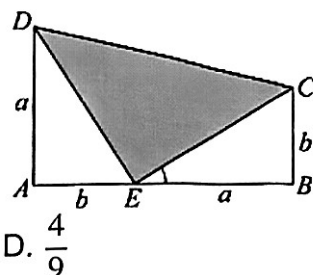
C. $\frac{2}{7}$

D. $\frac{5}{7}$

- 2 Eddie is playing darts, and the dart board is shown in the following figure. He will get the corresponding score when he throws the corresponding area. The circle's radius corresponding to 10 points is 1, and the circle's radius is increases by 1 for each outer layer. After throwing a dart, assuming Eddie does not miss the target, then what is the probability that Eddie gets 9 or 10 points? _____



- 3 In the right trapezoid $ABCD$ as shown in the figure, we can use the fact that "the sum of the areas of two congruent right triangles and one isosceles right triangle equals the area of the trapezoid" to give a simple and clear proof of the Pythagorean Theorem. This proof method is referred to as the "President's Proof." Given $a = 2$ and $b = 1$, a point is randomly chosen within the trapezoid $ABCD$. What is the probability that the point lies within the isosceles right triangle CDE (the shaded region)?



A. $\frac{2}{3}$

B. $\frac{5}{9}$

C. $\frac{1}{2}$

D. $\frac{4}{9}$

- 4 Randomly choose two numbers in the interval $[0, 2]$, then the probability that the smaller number of those two numbers is greater than $\frac{2}{3}$ is _____.

A. $\frac{1}{9}$

B. $\frac{1}{3}$

C. $\frac{4}{9}$

D. $\frac{2}{3}$

- Given that the x-intercept of the line $y = x + b$ lies within the interval $[-2, 3]$, what is the probability that the y-intercept b of the line is greater than 1?

A. $\frac{1}{5}$

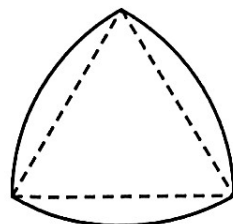
B. $\frac{2}{5}$

C. $\frac{3}{5}$

D. $\frac{4}{5}$

- 6 Take a rope with a length of 5 m, straighten it, and cut it in any position, then the probability of cutting both sections are not less than 2 m in length is _____ .

- 7 Using each vertex of an equilateral triangle as the center and the side length as the radius, draw an arc between the other two vertices. The three arcs together form a curved triangle known as the Reuleaux triangle. If a point is randomly selected from within the Reuleaux triangle, what is the probability that the point lies inside the original equilateral triangle?



- A. $\frac{2\pi - 3\sqrt{3}}{2(\pi - \sqrt{3})}$ B. $\frac{\sqrt{3}}{2(\pi - \sqrt{3})}$ C. $\frac{\sqrt{3}}{2(\pi + \sqrt{3})}$ D. $\frac{2\pi - 3\sqrt{3}}{2(\pi + \sqrt{3})}$

- 8 Given a square $ABCD$ with side length 2, a point is randomly selected from inside the square. What is the probability that the distance from the selected point to each of the four vertices A , B , C , and D is greater than 1?

- A. $\frac{\pi}{16}$ B. $\frac{\pi}{4}$ C. $\frac{3 - 2\sqrt{2}}{4}\pi$ D. $1 - \frac{\pi}{4}$

- 9 There is an isosceles right triangle that the length of the two equal sides is 2. Now take any point from one of the equal sides, then what is the probability that the distance from this point to the vertex of the right angle is less than 1?

A. $\frac{\pi}{8}$ B. $\frac{\pi}{4}$ C. $\frac{3\pi}{8}$ D. $\frac{\pi}{2}$ E. $\frac{\pi}{3}$

- 10 The school bus departs at 7 : 00, 8 : 00, and 8 : 30. If you arrive at the station at a random time between 7 : 50 and 8 : 30 to take the school bus, find the probability that you wait for no more than 10 minutes.

A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{2}{3}$ D. $\frac{3}{4}$